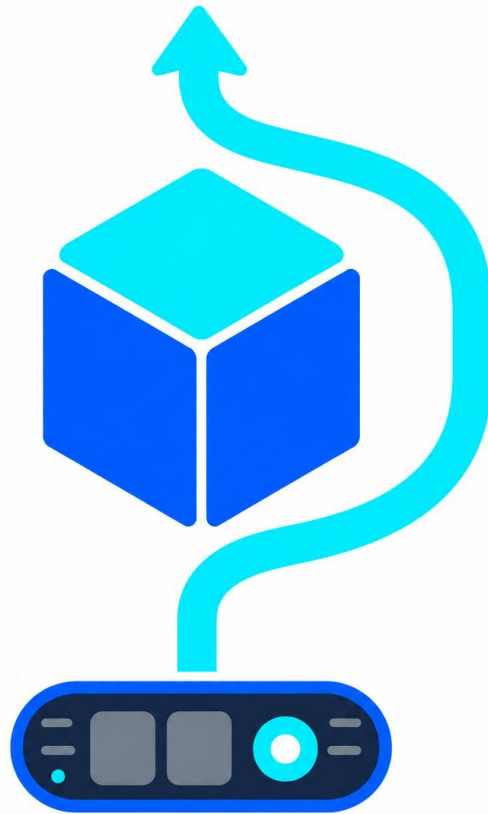




Obstacle Avoidance Solution

White Paper

3D sensing, zone status and obstacle geometry

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V0.1 · 2026-09-21

1 Solution overview

3D obstacle detection for mobile robots, providing input for slowing, stopping and planning around obstacles.

1.1 Shared capabilities and supported cameras

V1 and V2 share the same zone detection capabilities and algorithm performance. Both report Safe, Warning and Alarm. V2 adds obstacle bounding boxes and depth point-cloud output. Both support up to 20 templates for different operating conditions.

V1 / V2	Cameras	Capabilities and use
V1	S10 / S11	Zone status Provides input for slowing and stopping.
V2	S10 Pro	Zone status, plus two features obstacle bounding boxes and depth point-cloud output.

1.2 Applications



Warehouse robots

AGVs, AMRs and forklifts
Obstacles and aisle occupancy



Cleaning and delivery

Cleaning and service robots
Low objects and tight spaces



Close-range coverage

Near-body obstacles
Cover local blind spots



Warehouse and factory logistics

AGVs, AMRs and forklifts: detect loads, pallets and occupied aisles.



Commercial cleaning and service

Cover low obstacles, furniture and tight aisles, including turns and close passes.

MRDVS application images. The blue overlay illustrates coverage; it is not a detection-range specification.

System roles This solution provides perception and workstation configuration tools. The user supplies the controller and implements path planning, motion decisions and execution logic.

2 How detection works

2.1 From depth data to results



Depth points are evaluated against the active detection zones

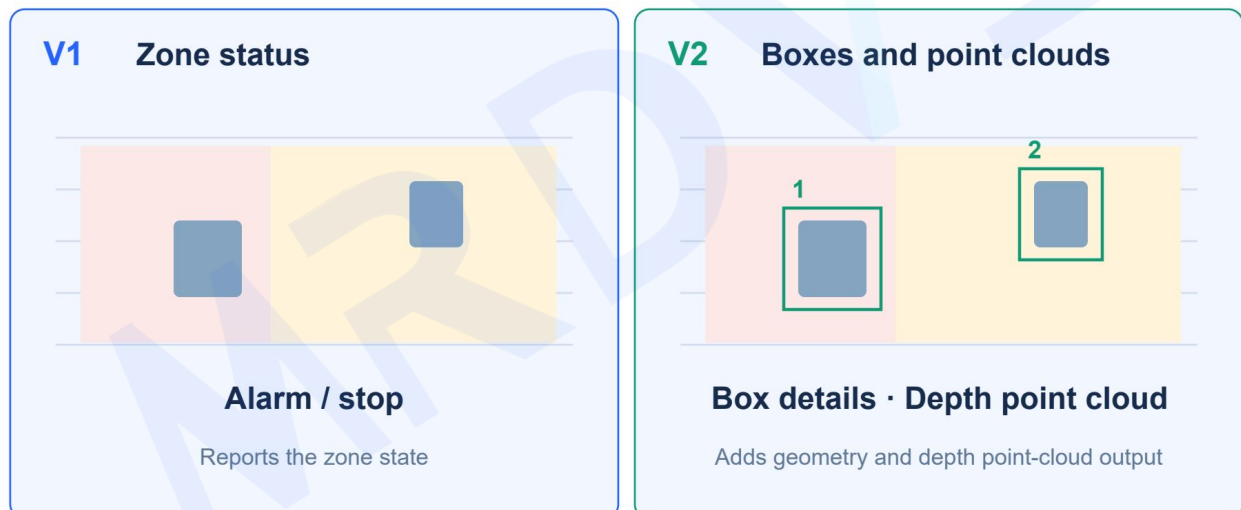
Depth and point clouds

The camera measures distance across the scene to form a 3D point cloud. Detection uses spatial position, not object color or category.

Alignment and zone checks

Mounting parameters align points to vehicle coordinates. Ground calibration aligns the floor; the active template defines which distances, widths and heights to evaluate.

2.2 Understanding the results



Both report zone status; V2 adds obstacle bounding boxes and depth point-cloud output

Zone status

Applies to the configured zones. A safe state does not establish that unseen or hidden space is clear.

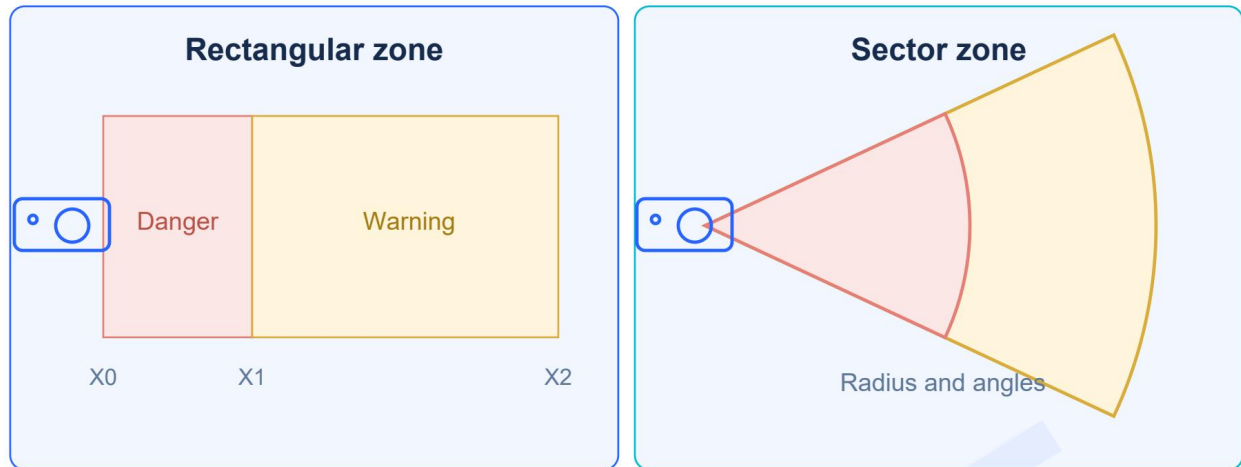
Perception output

The algorithm supplies perception data only. It does not plan paths or make motion decisions.

What is detected Obstacle geometry, without classifying objects as people, pallets or other categories.

3 Detection zones and output options

3.1 Zone shape and height



Top view: danger zone in red and warning zone in yellow; height limits apply to both

Rectangle

Distance, width and height limits for a straight corridor.

Sector

Radii, angles and height limits for a forward-facing arc.

3.2 Templates and applied settings

Up to 20 switchable templates

Store up to 20 zone configurations (IDs 0–19) and switch the active template as conditions change. Automatic switching requires integration with a user-provided controller.

Preview and detection

Preview changes before applying them. Detection uses the active settings on the camera.

3.3 V2 data and output settings

Data / setting	Purpose
Obstacle bounding boxes	Describe the space occupied by each detected obstacle for use by the user-provided controller.
Depth point cloud	Provide depth point-cloud data for processing in the user's system.
Simple output	Return only obstacle status and the bounding-box count. Turn this off to receive box details or point clouds.

Camera coverage Keep zones within the depth field of view and usable range. Larger zones cannot recover unseen or missing data.

4 Choosing a camera

Choose the output first, then check camera coverage and mounting. These are camera specifications, not guaranteed obstacle-detection performance.

Item	S10	S10 Pro	S11
Compatibility	V1	V2	V1
Depth FOV H × V	90° × 62° or 120° × 78°	61° × 90°	136° × 56°
Depth sensing range	0.3–8 m @ 90% 0.3–3 m @ 5%	0.3–17 m @ 90% 0.3–13 m @ 10%	Outdoor: 0.1–5 m 10%–90% reflectivity
Depth resolution	240 × 160	240 × 160	240 × 96
Frame rate (typ. / max.)	15 / 20 fps	10 / 10 fps	10 / 15 fps
Depth accuracy	≤3 cm	≤3 cm	≤1 cm
IP rating	IP54	IP67	Ethernet: IP54 Contact MRDVS for IP67
Operating temperature	−20–60°C	−20–60°C	−20–60°C

Specification conditions

FOV tolerance: ±3°. Percentages are surface reflectivity; 5% and 10% are different test conditions.

Usable detection range

Camera depth range does not guarantee obstacle detection at every distance. Validate real objects under the intended lighting and mounting conditions.

4.1 Matching the camera to the task

S10

Zone status and I/O; choose the FOV variant for the required coverage.

S10 Pro

Individual obstacle geometry and longer depth sensing range.

S11

Wide-angle, close-range coverage; check vertical coverage after mounting.

Selection checks Confirm the FOV, interface and firmware. Depth accuracy does not specify obstacle-size accuracy.

5 Camera limits in practice



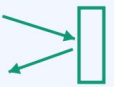
Field of view sets coverage

Keep zone edges within the depth field of view.
Mounting orientation changes width and height coverage.



Depth sensing has distance limits

Check both the minimum and maximum usable distance.
A larger configured zone does not extend camera range.

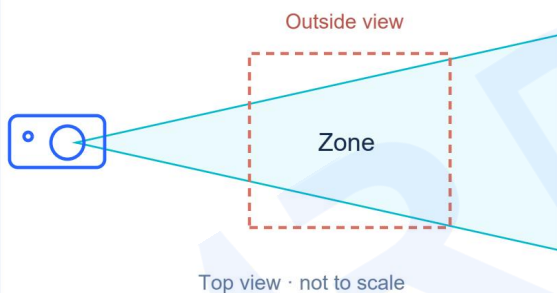


Reflectivity affects usable range

Reliable range can change with the target surface.
Test dark, mirror-like and transparent surfaces.

Check coverage, depth sensing range and object surfaces together

Zones must fit within the view



S10: reflectivity and distance

90%  8 m

5%  3 m

The object surface affects usable distance

Left: field-of-view concept. Right: S10 maximum depth sensing distance by surface reflectivity.

What to check on site

Mounting and coverage

Use the depth FOV, not the RGB image, to set zones.
Coverage changes with distance, mounting height, tilt and rotation.

Difficult objects

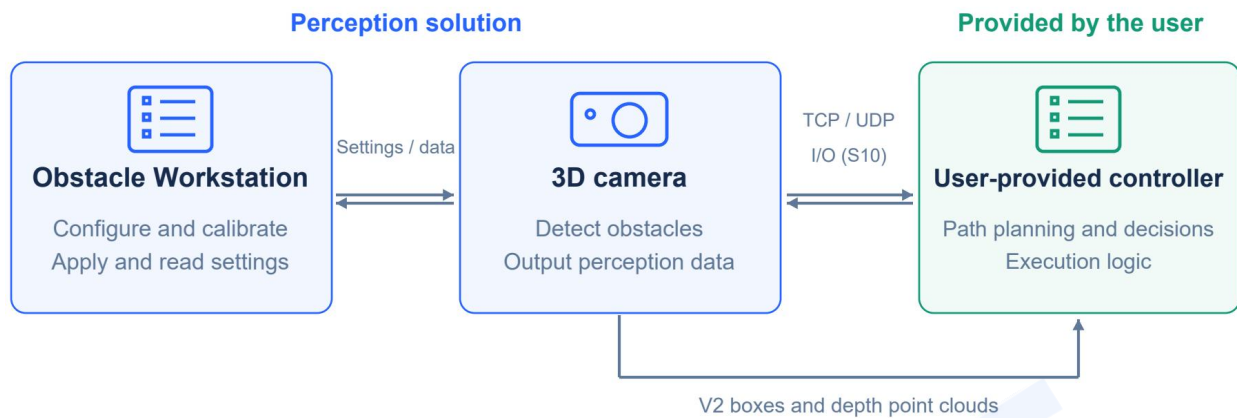
Test dark, mirror-like, transparent, thin, low and partly hidden objects for consistent detection.

Lighting

The 100 kLux rating does not replace testing under actual lighting and operating conditions.

6 System integration

The solution supplies perception. The user provides the controller and implements integration, path planning, motion decisions and execution logic.



The camera supplies perception data; the user-provided controller handles decisions and execution

Interface	Perception-side capability	User responsibility
TCP / UDP	Zone status, validity and active-template information; supports template selection.	Parse status packets and send template-selection commands as needed.
S10 I/O	Digital status outputs and template-selection inputs.	Connect the wiring and implement signal handling and control logic.

Interface scope Physical I/O is available on S10 only. TCP/UDP status packets do not contain bounding-box details or depth point clouds.

6.1 V2 bounding boxes and depth point clouds

V2 outputs obstacle counts, positions, dimensions and bounding boxes, plus depth point clouds for the user's system to process. These perception data contain no routes or motion commands.

6.2 Control and fault handling in the user system

Slowing and stopping

The camera reports warning and alarm states. The user-provided controller defines when to slow, stop and resume.

Invalid or missing signals

The user defines and implements the system response to invalid status, a lost connection or a data timeout.

End-to-end response

Vehicle response depends on sensing, communications, control and actuators together. It must be validated on the user's complete vehicle.

7 Deployment and validation

Define the targets, site conditions and mounting. Calibrate, configure and test the system, and agree on the perception deliverables and the user's integration responsibilities.

7.1 Application and deployment requirements

Target objects

Specify object sizes, surfaces, distances and occlusion. Include low, thin and low-reflectivity objects.

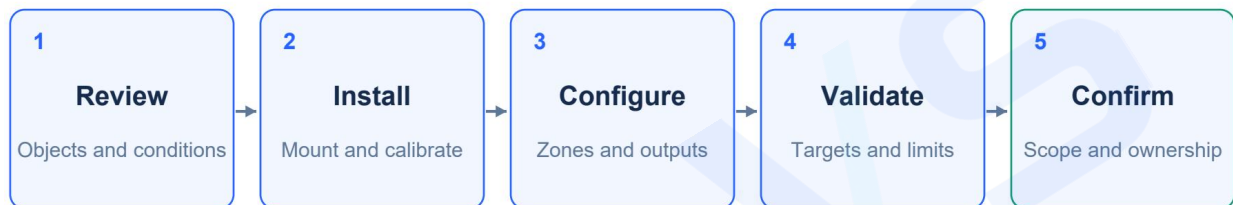
Operating conditions

Specify indoor or outdoor use, lighting, dust and floor conditions. Assess detection consistency on site.

Mounting and interfaces

Check mounting space, orientation and coverage, plus camera compatibility, required outputs and controller interfaces.

7.2 Deployment stages



Stage	Purpose and main tasks
Mount and calibrate	Establish the mounting position and align the point cloud with the vehicle and floor so zone directions and heights have a consistent reference.
Configure zones and outputs	Set zones and templates for the vehicle, route and target objects. Select the required outputs and apply the settings to the camera.
Test the scene and vehicle	Check empty zones, representative obstacles and zone boundaries. The user integrates the controller and verifies the complete vehicle response.

7.3 Delivery and validation boundaries

Scope	Project boundary
Provided by this solution	Zone-status detection; V2 adds bounding boxes and depth point clouds. The workstation configures perception and displays results. Path planning and motion control are not included.
Implemented by the user	The controller, communications integration, path planning, motion decisions, execution logic and vehicle-response validation.
Established through evaluation	Validate detectability, usable distance, coverage and consistency with real targets and mounting. Agree on capability and responsibilities; reassess when targets, mounting or conditions change.

Near-ground detection Detecting low objects and suppressing floor noise involves a trade-off. Establish the usable height range for the installation and site conditions.

8 Frequently asked questions

Start with the required output, then confirm coverage and vehicle integration.

01 How do V1 and V2 differ?

Both have the same zone detection capabilities and algorithm performance. V1 reports zone status; V2 adds obstacle bounding boxes and depth point-cloud output. Choose based on the required output and camera coverage.

02 Does the camera drive around obstacles?

No. The solution provides perception only. The user supplies the controller and implements path planning, motion decisions and execution logic. V2 supplies bounding boxes and depth point clouds as input.

03 Does it classify people or pallets?

The solution detects occupied space; it does not label objects as people or pallets. V2 separates detected obstacles and reports their geometry.

04 What about dark or reflective objects?

Detection cannot be guaranteed from a material label alone. Low reflectivity affects usable range. Test dark, transparent, mirror-like, thin and partly hidden objects at the intended distances and angles.

05 Can I use the full specified range?

Not necessarily. Maximum range assumes specified surfaces and conditions. FOV, mounting and occlusion also limit coverage. Validate the usable area and set zones for the complete vehicle response.

06 Is calibration and testing still needed?

Yes. Calibration establishes the spatial alignment; site testing confirms usable coverage. Include empty zones, low objects and detection boundaries. Reassess after mounting, parameter or firmware changes.

07 Can it detect low obstacles?

Detection depends on the camera, mounting and detection height settings. A lower height limit set too high can exclude low objects; setting it too low can introduce floor interference. Test with real targets and an empty zone to confirm performance.

Revision history and support

Revision history

Version	Date	Revision
V0.1	2026-09-21	Initial release covering capabilities, camera selection, operating limits and integration.

Technical support

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Knowledge Community: <https://docs.mrdvs.com/en>

LinkedIn: <https://www.linkedin.com/company/mrdvs>



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